

2. Data circuits. Dedicated local or long-distance point-to-point or point-to-multipoint data circuits at different speeds, namely, $n \times 64$ kbps and up to 2Mbps can be offered for different bandwidth needs of the customer. MLLN offers flexibility of providing leased circuits with speeds of $n \times 64$ kbps and up to 2Mbps with differential time-dependent bandwidth provisioning.

3. Private data network. More than one local or long-distance leased circuits can be provided such that data from one leased circuit can be transferred automatically to another leased circuit for the same subscriber.

4. International leased circuits. International long-distance leased circuits can be offered for business across the globe, which are useful for Internet leased lines and international private leased circuits (IPLCs).

The MLLN also supports enhanced features such as corporate hi-speed Internet access, EPABX interconnection, EPABX remote extension (EPABX of one city can be connected to EPABX of another city), ISDN (integrated services digital network) line extension, virtual private network (VPN), local area network (LAN) interconnection (LAN of one city can be connected to LAN of another city) and extension of VPN to customer through MLLN. A typical MLLN system connecting various facilities of a customer is shown in Fig. 1.

Salient features of MLLN

The MLLN service is specially designed for having effective control and monitoring on the leased line so that the downtime is minimised and the circuit efficiency is increased.

MLLN mainly deals with data circuits ranging from 64kbps to 2048kbps ($n \times 64$ kbps). One of the major attractions of MLLN is its ability to provide differential time-dependent bandwidth on demand basis to the customers. For example, it is possible to provision 1024kbps MLLN circuit for 16 hours a day and 512kbps for remaining 8 hours of the day, as per the requirement of the customer.

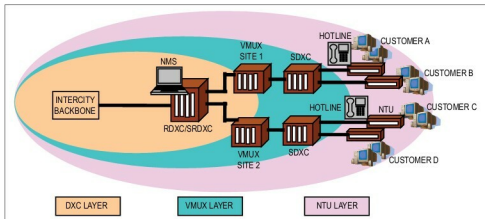


Fig. 2: MLLN architecture

Another great thing about MLLN is its very efficient NMS that can proactively maintain the circuit without waiting for customers to book a complaint. NMS provides features such as bandwidth management, alternate or back-up transmission routing, powerful diagnostics and maintenance tools and self-repair tools.

MLLN-NMS also provides periodic performance report which is useful in providing high-speed leased lines with improved QoS, high availability and reliability to the business and good service to existing customers.

The network management system also supports service provisioning, network optimisation and planning and service monitoring. The system offers features such as end-to-end circuit creation and monitoring, software loop test to check connectivity of various network elements and fault isolation and software programmability of customer end equipment.

MLLNs offer great amount of security because media is not shared and is exclusively dedicated for a particular subscriber. With MLLN, lead time is very low for provisioning of a new leased line. Its modular system and new application can be implemented very fast by simply adding or plugging the units.

MLLN architecture

MLLN network is a three-tier structure and comprises network elements such as digital cross connects (DXC), versatile multiplexer (VMUX), network termination units (NTU) and NMS (Fig. 2).

Stage 1 comprises NMS, regional DXC (RDXC) and/or sub-regional DXC (SRDXC), billing servers, database servers, etc. At this stage, all the network management functions are done from the central location. This stage provides connectivity to second-stage nodes and provides traffic aggregation.

Second stage comprises SRDXC, SSDXC (secondary switching DXC) and VMUX, and is located in major cities where demand for leased line is high. This stage provides connectivity to third stage and performs leased line traffic aggregation.

Third stage comprises VMUX and NTU and is located in small cities/towns where leased line demand is lower (approximately 20). This stage provides leased line traffic aggregation.

The media for interconnecting various network elements and extending line up to customer end may be optical fibre, copper wire, radio, microwave transmission or a combination of these.

MLLN has to provide high reliability service and is supposed to obtain efficiency greater than 99.5 per cent. Therefore all the interconnections of different network elements are provided as rings, wherever available, so that an alternate circuit path can be automatically used for routing the traffic in case of main route failure. In long-distance network, links between the same stations can be split into alternate physical path of rings to the extent feasible. This also saves on-port capacity required for providing alter-